following pages, I give an account of what we have understood about time and the paths that are being followed in our search to understand it better, as well as an account of what we have yet to understand and what it seems to me that we are just beginning to glimpse.

Why do we remember the past and not the future? Do we exist in time, or does time exist in us? What does it really mean to say that time "passes"? What ties time to our nature as persons, to our subjectivity?

What am I listening to when I listen to the passing of time?

This book is divided into three unequal parts. In the first, I summarize what modern physics has understood about time. It is like holding a snowflake in your hands: gradually, as you study it, it melts between your fingers and vanishes. We conventionally think of time as something simple and fundamental that flows uniformly, independently from everything else, from the past to the future, measured by clocks and watches. In the course of time, the events of the universe succeed each other in an orderly way: pasts, presents, futures. The past is fixed, the future open. . . . And yet all of this has turned out to be false.

One after another, the characteristic features of time have proved to be approximations, mistakes determined by our perspective, just like the flatness of the Earth or the revolving of the sun. The growth of our knowledge has led to a slow disintegration of our notion of time. What we call "time" is a complex collection of structures,[2] of layers. Under increasing scrutiny, in ever greater depth, time has lost layers one after another, piece by piece. The first part of this book gives an account of this crumbling of time.

The second part describes what we have been left with: an empty, windswept landscape almost devoid of all trace of temporality. A strange, alien world that is nevertheless still the one to which we belong. It is like arriving in the high mountains, where there is nothing but snow, rocks, and sky. Or like it must have been for Armstrong and Aldrin when venturing onto the motionless sand of the moon. A world stripped to its essence, glittering with an arid and troubling beauty. The physics on which I work—quantum gravity—is an attempt to understand and lend coherent meaning to this extreme and beautiful landscape. To the world without time.

The third part of the book is the most difficult, but also the most vital and the one that most closely involves us. In a world without time, there must still be something that gives rise to the time that we are accustomed to, with its order, with its past that is different from the future, with its smooth flowing. Somehow, our time must emerge around us, at least *for* us and at our scale.[3]

This is the return journey, back toward the time lost in the first part of the book when pursuing the elementary grammar of the world. As in a crime